



Eat the food you love. **Know what it does to you.**

An on-device AI nutrition and metabolic-health coach built for Indian meals.

SHAKTI Hackathon · Obesity Control & Medical Fitness · AI-Agentic

Live: thalisense.vercel.app

github.com/andringodson/ThaliSense

PROBLEM

India's diabetes and obesity crisis is cooked in everyday kitchens, and our tools don't speak its language.

101 M

Indians living with diabetes

136 M

more with prediabetes, mostly undiagnosed

BMI 23

is where overweight starts for Asian Indians, not the global 25

- Calorie apps are built for Western diets. They don't know pesarattu, bisi bele bath or kadhi, or what "do roti aur dahi" means.
- Risk is mis-measured. South Asians get diabetes at lower weight, so global BMI tells at-risk people they're fine.
- "Eat less rice" doesn't stick, dietitians are unaffordable, and people won't upload health data to a server.

PROPOSED SOLUTION

A coach that understands Indian food, measures risk the Indian way and plans meals you'll actually eat.

1 • Log

Type it your way ("rendu dosai", "2 roti, dal") or snap a photo. On-device AI recognises 108 Indian dishes.

2 • Understand

Indian Diabetes Risk Score, Asian BMI cut-offs, waist-to-height ratio and personal calorie and protein targets.

3 • Coach

An agent finds what matters most and swaps foods you already eat, showing the weekly impact of each swap.

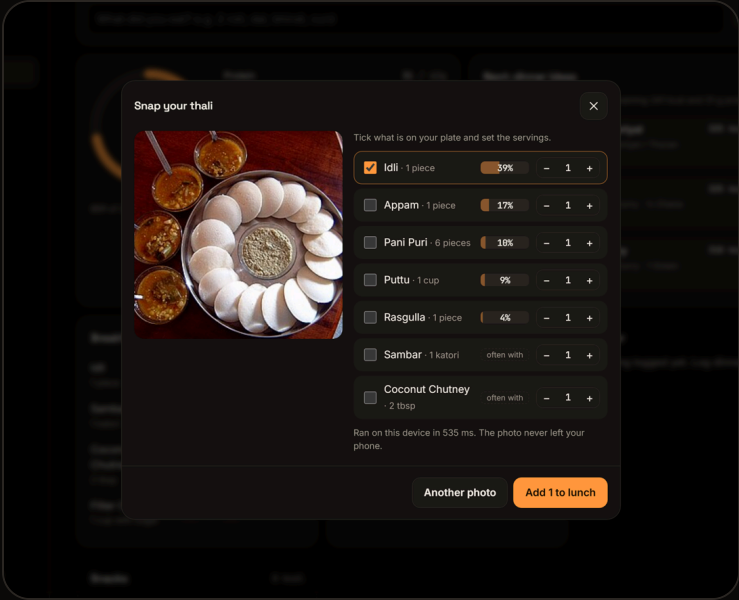
4 • Plan

A self-checked 7-day regional meal plan for your diet and ₹/day budget. Log any meal with one tap.

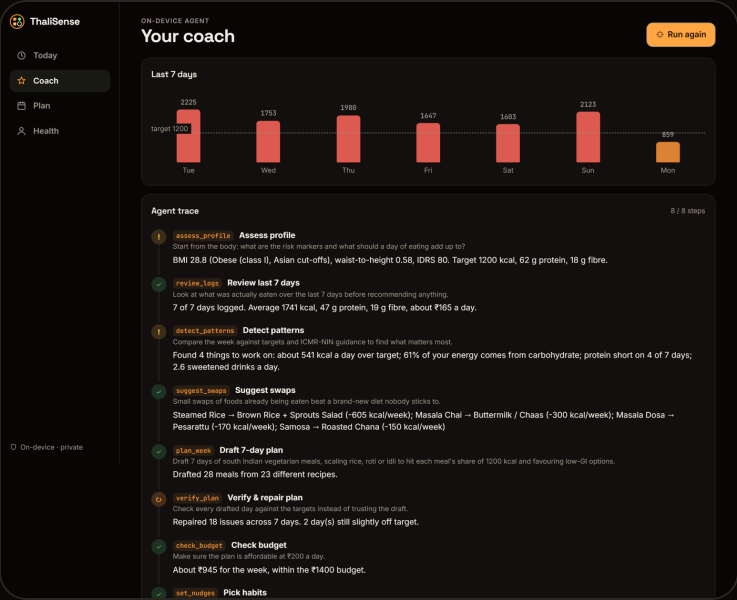
Free. No sign-up. **Nothing leaves your phone:** the AI runs inside the browser.

WORKING MVP · LIVE TODAY

From a photo of your thali to a week of better meals.



Snap: dish recognised on the device in about 1 second, with usual sides suggested.

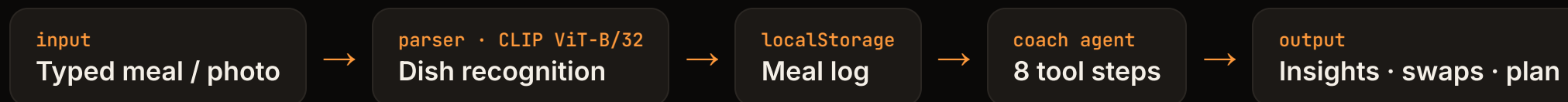


Coach: an 8-step agent trace you can follow, plus insights and swaps.



Plan: 7 regional days within budget, with notes on what the coach adjusted.

Real AI, running entirely on the user's phone.



Vision: zero-shot CLIP in the browser

Transformers.js on ONNX Runtime Web (WASM) in a Web Worker. Dish text embeddings are precomputed at build time with prompt ensembling, so the phone downloads only the 89 MB int8 vision encoder once and classifies a photo in about 1 second.

Agent: plan, verify, repair

A tool-based planner: **assess** → **review** → **detect** → **swap** → **plan** → **verify** → **budget** → **habits**. It scores 60 real meal templates on protein, fibre, glycaemic load, cost and region, scales the staples, then checks every day against the targets and fixes the misses.

React 19 · TypeScript · Vite · Transformers.js · ONNX Runtime Web · Vitest · Playwright · GitHub Actions · Vercel · no backend, no paid APIs

IMPLEMENTATION APPROACH

Built from scratch for SHAKTI, grounded in Indian clinical science.

How we built it

- Food data: 108 dishes from IFCT 2017 (ICMR-NIN) with serving sizes and cost
- Risk: MDRF IDRS, Asian BMI consensus, Mifflin-St Jeor, ICMR-NIN 2024
- Vision: CLIP embeddings precomputed at build time; tested on 30 real photos
- Agent: template scoring with a verify-and-repair loop; 15 unit tests
- Shipped: CI on every push, auto-deploy to Vercel

Status & what is new

Everything shown was built during the hackathon window. No pre-built project code was reused; open-source libraries (React, Transformers.js) and the public CLIP model are credited.

Typed logging, 108 dishes	Live
On-device photo recognition	Live
Risk scores and targets	Live
Coach agent and 7-day plan	Live
Multi-dish thali, voice in Indian languages	Next

EXPECTED IMPACT

Small swaps, caught early, at national scale.

~1,200

kcal a week saved by the swaps the coach finds in the demo week, about 0.7 kg of fat a month, with more protein and fibre.

60+

IDRS score triggers a clear "get tested" prompt, catching undiagnosed diabetes earlier.

0

bytes of health data uploaded. Privacy builds trust, especially for women and older users.

- Fights obesity with the food people already eat, not a foreign diet they'll quit in a week.
- Inclusive by design: vegetarian-aware, budget-aware (₹200 a day by default), regional, and runs on low-end phones.
- Scales at near-zero cost: static hosting, on-device AI, works offline.

FUTURE SCOPE

Where ThaliSense goes next.

- Multi-dish thali detection: segment the plate, then recognise each katori.
- Voice logging in Hindi, Tamil, Bengali and Marathi.
- Progress tracking: weight, waist and HbA1c trends, with goals.
- ASHA-worker mode: offline community IDRS screening and referrals.
- Wearable and glucometer import for steps and glucose response.
- Optional LLM chat, grounded in the agent's trace, with the user's own key.



Try it now.

Open the link on any phone and press **Try with a demo week.**

Live app

thalisense.vercel.app

Source code

github.com/andringodson/ThaliSense